ACYCLIC RELATION

E Id

ORDERING	NOT REFL. ✗	REFL. WEAKLY ANT TRANS. ✓
STRICT ORDERING	ANTIREFL. ✗ XTRANS.	NOT ANTIREFL. ✗

$\longleftrightarrow$

$(\emptyset, \emptyset)$

$\emptyset = \emptyset$

$\updownarrow$

ANTISYMM: $x \in y \rightarrow \neg y \in x$

FROM FOUNDATION AX

$\phi \in \{\phi\} \in \{\{\phi\}\}$

$\phi \notin \{\{\phi\}\}$

$\subseteq$

- X IS <u>BOUNDED FROM ABOVE</u> (BELOW) IN A IF THERE EXISTS A MAJORANT OF X IN A

- X IS A <u>LOWER SET</u> (CLASS) (DOLNÍ MNOŽINA) IN A IF $(\forall x \in X)(\forall y \in A)(y \leq_R x \rightarrow y \in X)$
- $x \in A$ THEN $(\leftarrow, x]$ IS $\{y;\ y \in A \wedge y \leq_R x\}$. A <u>PRINCIPAL IDEAL</u> (HLAVNÍ IDEÁL) DETERMINED BY x

$\boxed{\text{OBSERVATION}}$ R ORDERING ON A, THEN FOR ARBITRARY $x, y \in A$ WE HAVE $x \leq_R y \longleftrightarrow (\leftarrow, x] \subseteq (\leftarrow, y]$

$\boxed{\text{REMARK}}$ CONSTRUCTION OF $\mathbb{R}$ FROM $\mathbb{Q}$ ... <u>DEDEKIND'S CUTS</u>: $X \subseteq \mathbb{Q}$, X IS A LOWER SET AND IF $\sup X$ EXISTS THEN $\sup X \in X$

$(\mathbb{Q} \cap (-\infty, 1)$ IS NOT A DED. CUT BUT $\mathbb{Q} \cap (-\infty, 1)$ IS A DED. CUT

$\mathbb{Q} \cap (-\infty, \sqrt{2})$ IS A DED. CUT $\{x;\ x \in \mathbb{Q};\ x^2 \leq 2 \vee x \leq 0\}$

DEF AN ORDERING R ON A CLASS A IS A <u>WELL-ORDERING</u> (DOBRÉ USPOŘÁDÁNÍ) IF

EVERY <u>NONEMPTY</u> SUBSET M OF A HAS A <u>SMALLEST</u> ELEMENT WITH RESPECT TO R.

STRICT WELL-ORDERING "$\leq$ − Id"

EX. WRITE THIS DEFINITION USING A FORMULA

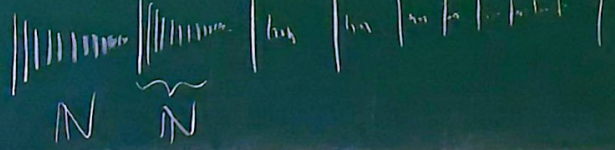
OBSERVATION: WELL-ORDERING IS A HEREDITARY CONDITION ($B \subseteq A$, THEN R IS A WELL-ORDERING ON B)

EVERY WELL-ORDERING IS A LINEAR ORDERING

$(\mathbb{N}, \leq)$

$(\{0, 1, 2, \dots m\}, \leq)$

$(\mathbb{N} \times \mathbb{N}$, LEXICOGR. ORDERING$)$



$(\mathbb{Z}, \leq)$ $(\mathbb{Q}, \leq)$

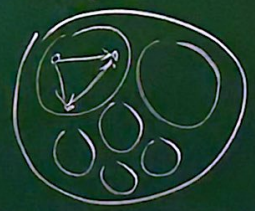
$(\mathbb{R} \leq)$

$\{x, y\}$

EX FIND SOME SETS ON WHICH E ($\in$) IS A STRICT WELL-ORDERING

EQUIVALENCE

EQ. CLASSES $\longleftrightarrow$ PARTITIONS



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